CHAITANYA PRABHU

BACKEND ENGINEER

New Delhi, India

Backend engineer building high-performance, fault-tolerant services and APIs — from the request parser up to the data layer. I write servers and frameworks from the socket up in TypeScript and C, design data layers in PostgreSQL and Redis, and ship with Docker and Kubernetes on AWS. Focused on distributed systems and the reliability problems — backpressure, failover, consistency — that decide whether a service holds up under load.

EDUCATION

IIIT Hyderabad — M.Tech, Computer Science — 2025 – PRESENT

CGPA 7.44 / 10 (semester 1) · Coursework: Advanced Operating Systems, Software System Development, Data Structures & Algorithms, System & Network Security.

NIT Silchar — B.Tech, Civil Engineering — 2020 – 2024

GPA 7.4 / 10

TECHNICAL SKILLS

LANGUAGES	TypeScript / JavaScript, C, C++, Python, SQL, Bash
BACKEND & WEB	Node.js, FastAPI, Next.js, React, REST, GraphQL, WebSockets, JWT / auth
DATA & INFRA	PostgreSQL, Redis, MySQL, MongoDB, Docker, Kubernetes, AWS
FOCUS	Distributed systems, API & framework design, performance engineering, microservices, RAG / LLM pipelines

PROJECTS

forge-http — HTTP/1.1 Server Framework — TYPESCRIPT · 2026

- Production-grade HTTP/1.1 server framework written from scratch on raw TCP sockets with zero runtime dependencies; a byte-level incremental parser reassembles requests across packets and stays strict against request smuggling — rejecting conflicting Transfer-Encoding / Content-Length, obsolete line folding, and oversized headers.
- Radix-trie router with path params and wildcards over an onion-model async middleware engine; keep-alive, pipelining, chunked transfer, range requests, ETags / 304s, gzip / brotli / deflate, WebSockets (RFC 6455) and SSE; JWT with algorithm pinning and constant-time verify, CORS, sliding-window rate limiting, Prometheus metrics, graceful shutdown — ~10k req/s at p99 ≈ 8 ms.

TYPESCRIPT · RAW TCP SOCKETS · HTTP/1.1 · WEBSOCKETS · TLS · ZERO DEPS

ephemeral-secret — Self-Destructing Secret Sharing — TYPESCRIPT · 2026

- One-time secret-sharing service over HTTPS: secrets are encrypted client-side with AES-256-GCM under an HKDF-derived key that lives only in the URL fragment, so the key never reaches the server — it stores ciphertext it cannot read.
- One-time retrieval via atomic Redis GETDEL (no read-then-delete race), TTL-based auto-expiry, and authenticated encryption that hard-fails on any tampering. Zero server-side key storage.

TYPESCRIPT · AES-256-GCM · HKDF · REDIS · ZERO SERVER-SIDE KEY STORAGE

raft-kv — Fault-Tolerant Distributed Key-Value Store — C · 2026

- Three-node distributed key-value store implementing the Raft consensus algorithm in pure C over raw POSIX sockets with zero dependencies — leader election, log replication, and crash recovery; an acknowledged write survives any single-node failure, including a hard SIGKILL, and a rejoining node replays its log to catch up.
- Four-thread node over a single mutex-protected state machine; log appended with O_DSYNC for WAL-grade durability; hand-rolled binary wire protocol framed through recv_all. Validated by a chaos suite — random kills, iptables partitions, failover — clean under TSan / ASan / UBSan.

C · RAFT CONSENSUS · POSIX SOCKETS · O_DSYNC DURABILITY · PTHREADS

EXPERIENCE

Engineer Intern — Deendayal Port Authority — MAY – NOV 2023

Gandhidham, Gujarat · On-site

- Maintained the relational cargo-logging database for Oil Jetty No. 8 (liquid-bulk tanker discharge) — wrote SQL and PL/SQL to record and reconcile consignment and vessel data, enforcing validation and integrity across high-volume daily entries.
- Built a structured logging and audit trail of cargo events through discharge and produced tracking reports, giving operations real-time visibility into throughput and surfacing discrepancies early.

ACHIEVEMENTS

- **GATE 2025** CS & IT — 59.62 / 100 · **GATE 2024** DS & AI — 62.67 / 100 · **JEE Main 2020** — 98.28 percentile
- Winner, All-India Inter-NIT Chess (2022) · Runner-up, All-India Inter-NIT Football (2023)
- Head, Publication Team — Technoesis 2023, NIT Silchar